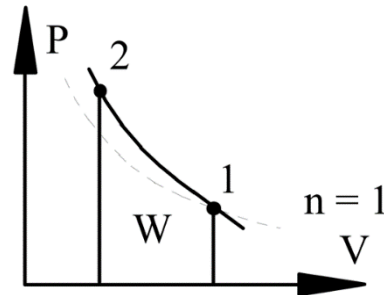


A piston cylinder contains 0.1 kg air at 100 kPa, 400 K which goes through a polytropic compression process with $n = 1.3$ to a pressure of 300 kPa. How much work has the air done in the process?

Process: $Pv^n = \text{Const.}$

Work integrated term:

$${}_1W_2 = \frac{1}{1-n} (P_2 V_2 - P_1 V_1)$$



Solution:

Process: $Pv^n = \text{Const.}$

$$\begin{aligned} T_2 &= T_1 (P_2 V_2 / P_1 V_1) = T_1 (P_2 / P_1)(P_1 / P_2)^{1/n} \\ &= 400 \times (300/100)^{(1 - 1/1.3)} = 515.4 \text{ K} \end{aligned}$$

Work term is already integrated giving Eq.4.4

$$\begin{aligned} {}_1W_2 &= \frac{1}{1-n} (P_2 V_2 - P_1 V_1) = \frac{mR}{1-n} (T_2 - T_1) \quad \text{Since Ideal gas,} \\ &= \frac{0.1 \times 0.287}{1 - 1.3} \times (515.4 - 400) = \mathbf{-11 \text{ kJ}} \end{aligned}$$